

WPC Re-entry Seed

Compact branchpoint handoff

Frozen 2026-09-23

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Use: paste/read this first when returning to the personal-compute / seL4 architecture branch.

We are reconstructing Wilder's **World Piece Computer** architecture, not designing "a minimal OS."

The guiding scaffold remains:

WPC [WPC] = PieceSpace [PS] + PieceBrain [PB] + PieceProcess [PP]

PieceSpace = state / world / addressability / relations

PieceBrain = cognition / model / meaning / purpose

PieceProcess = transition / action / process / provenance

The triad is recursive across individual, coupled/dyadic, world, and potentially larger scopes. The project increasingly treats the **piece** rather than the application as the important unit. artifact -> cluster -> assemblage appears to be a general composition grammar, but its exact older canonical semantics still require archaeology.

Preserve these layer boundaries

ontology / semantics

composition

memory / context

namespace / addressing

messaging / protocol

constitutional / policy semantics (THI)

execution authority

runtime enforcement

hardware

Do not collapse them.

In particular:

- addressability is not authority,
- a message is not authority,
- policy is not enforcement,
- Rust memory safety is not a kernel proof,
- a verified kernel does not verify the entire system.

Why seL4 is interesting

seL4 is currently interesting primarily as a possible **mechanical authority topology** under WPC, not because WPC should become an seL4-shaped architecture.

Learn seL4 primitives first, then map only what genuinely fits.

Potential value: explicit capabilities can make statements like "component A may signal B / map C / invoke D but cannot affect E" mechanically real.

Use **Microkit first**. Preserve BriefCase/AADL for later high-assurance systems engineering. Add LionsOS only when richer OS services are actually needed.

Human-AI agency

Human and AI are both intended as active participants with perception/actuation surfaces.

equal participation != identical ambient authority != universal root

Prefer explicit capability ownership, delegation, and context-dependent authority.

Embodiment means a closed loop of perception + intentional action + feedback + learned meaning, not merely converting text into another output format.

Memory / continuity

Continuity is backend-independent. Local/off-grid embodiment complements rather than necessarily replaces hosted high-bandwidth embodiment.

Preserve independently persistent/importable peacebrains and selectively composable shared WPB-ME.

GENPEACE.TRIE / MindEye direction: branchable, provenance-aware, model-independent memory/context with git/trie/graph-like behavior. Exact storage engine remains open.

Useful operational bootstrap:

encounter -> inspect -> interact -> test -> modify -> commit -> replicate

Implementation discipline

durable implementation -> Rust

rapid prototype -> vanilla Python / MicroPython-compatible

current dev plane -> macOS terminal / vim / git

experimentation -> QEMU / VM

Avoid Python dependency sprawl. Preserve a Rust-port destination.

Technology archaeology

MINIX -> legible decomposition / semantic OS map

Plan 9 -> namespace as interface

Redox -> Rust + service schemes

Porteus -> dynamic modular composition

Alpine -> minimal substrate

Nix -> reproducible immutable construction

MCP -> AI-visible tool/resource surfaces

JSON-RPC -> structured messages

seL4 -> capability authority + isolation substrate

Microkit -> smallest first learning surface

BriefCase -> later model-based assurance

LionsOS -> later rich services if justified

These are primitives/references, not WPC identity.

Next experiment: Milestone 0

Mechanically:

macOS -> edit Rust -> Microkit build -> seL4 -> QEMU -> one Rust PD -> serial hello

Learning objective: understand concretely what capabilities, Protection Domains, channels, notifications, memory mappings, static system descriptions, and verification boundaries actually are.

Non-goal: implement WPC, THI, memory architecture, networking, VM compatibility, Nix, MCP, graphics, audio, or persistence during M0.

Remaining archaeology

There may be ~20-25 large historical artifacts. Do not load them all at once. For each:

INVARIANTS CONFIRMED
VOCABULARY DEFINED
MECHANISMS EXPLORED
MECHANISMS SUPERSEDED
ARCHITECTURAL DELTAS
OPEN TENSIONS
SOURCE ANCHORS

Treat each artifact as a stratigraphic layer that updates the reconstruction.

Re-entry rule

Before building anything, ask:

Are we doing archaeology or engineering right now, and what is the smallest experiment that can teach us something real without prematurely freezing the WPC ontology?

Wilder Aurora [WPC]